

Composites of Hydroxyethyl Cellulose and Calcium Sulfate Whiskers Enable Durable Thermoelectric Devices with High Flame Retardancy

Jizhen Huang, Shaolu Qi, Zhiyong Xu,* Jiao Ji, Wenbo Zhao, Jianquan Hu, and Yuxin Liu*

Cite This: *ACS Appl. Polym. Mater.* 2025, 7, 10224–10234

Read Online

ACCESS |



Metrics & More



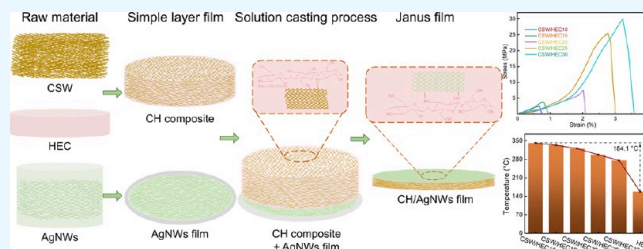
Article Recommendations



Supporting Information

ABSTRACT: The high-temperature-resistant composite matrix material, derived from one-dimensional inorganic nano materials rather than the traditional organic polymer matrices used in flexible devices, can significantly broaden the application range of wearable devices, particularly in high-temperature or open-flame environments. In this study, flame-retardant and high-temperature-resistant composite nanofilms were prepared by solution casting with hydroxyethyl cellulose (HEC) bonded to calcium sulfate whiskers (CSW). When the HEC content reached 20%, the composite films exhibited a tensile strength of 8.20 MPa and a limiting oxygen index exceeding 44.5%. Molecular dynamics simulations indicated that the electrostatic and van der Waals interactions between HEC and CSW were strongest in the CSW/HEC20 sample with a binding energy of 9227.23 kJ/mol. Furthermore, when the CSW/HEC20 film was integrated with silver nanowires, a flexible electrothermal converter was successfully fabricated, providing a base material for high-temperature-resistant electronic devices.

KEYWORDS: calcium sulfate, nanowhisker, cellulose, fire resistance, thermoelectric conversion



1. INTRODUCTION

The rapid development of flexible electronic devices has led to the widespread application of flexible electrothermal converters in wearable technology, medical monitoring, and specialized heating environments.^{1,2} These flexible electronic sensors consist of heating and matrix materials, which are predominantly organic, such as polyethylene terephthalate (PET), polydimethylsiloxane (PDMS), chitosan, and cellulose.^{3–5} While organic matrix materials offer good flexibility and toughness, they exhibit poor temperature resistance and flame retardancy, limiting the application of flexible electrothermal converters in high-temperature environments and precise temperature control.^{6,7} Calcium sulfate whiskers (CSW), a typical monoclinic inorganic material, possess advantages such as noncombustibility, high temperature resistance, and non-toxicity.⁸ Recent studies have demonstrated that hydrothermal synthesis combined with calcination enables the preparation of calcium sulfate anhydrous whiskers (CSAW) with an average aspect ratio exceeding 25, which can be embedded into polymer matrices ((e.g., poly(vinyl alcohol)/montmorillonite (PVA/MMT)) to form a “columnar-honeycomb” structure, significantly enhancing mechanical strength and thermal stability.⁹ Employing CSW as the substrate material for flexible sensors significantly enhances thermal resistance and flame retardancy ratings, thereby substantially expanding the application scope of these devices.¹⁰ However, calcium sulfate nanocrystalline films are brittle and prone to fracture upon impact, necessitating the incorporation of polymer materials to

form composite structures.¹¹ Zhang et al. and Cong et al. employed CSW to enhance the mechanical properties of PVA and polypropylene (PP), but the temperature resistance of the composites remained insufficient.^{12,13}

Hydroxyethyl cellulose (HEC), a typical cellulose derivative, offers advantages such as nontoxicity, strong adhesion, and excellent film-forming properties.¹⁴ Huang et al. employed HEC as a cross-linker and dispersant to incorporate boron nitride nanosheets (BNNS). At a BNNS content of 20%, the composite film exhibited a tensile strength of 34.0 MPa and thermal conductivity of $0.234 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$.¹⁵ However, due to the limited binding sites between the two-dimensional inorganic nanosheets and HEC, the mechanical properties decreased when the BNNS content exceeded 20%. When the BNNS content was increased from 20% to 30%, the mechanical properties of the HEC/BNNS composite films improved, suggesting that the bonding between the inorganic nanosheets and the polymer may be enhanced.¹⁶ This is consistent with the conclusion that HEC hydroxyl groups can form mixed hydrogen bonding networks with organic/inorganic components to optimize interfacial interaction.¹⁷

Received: June 4, 2025

Revised: July 9, 2025

Accepted: July 9, 2025

Published: July 18, 2025



Understanding the binding properties between HEC and inorganic materials has thus become a focal point of research. Molecular dynamics (MD) simulation serves as a powerful methodology for probing the structural dynamics, deformation, destruction, and interfacial interaction of materials on the atomic scale, thus facilitating the microscopic insight that otherwise is often inaccessible by experimental techniques.¹⁸

In composite materials containing both organic and inorganic components, the organic substances are typically dispersed and bonded to the inorganic materials.¹⁹ The mechanical properties, insulation, thermal conductivity, and other characteristics of the composite depend on the size compatibility between the two materials and the strength of interfacial bonding. For example, incorporating cellulose-derived carbon dots (CDs) or crystalline cellulose nanofibers (CNFs) into hydrogel matrices has been shown to enhance structural integrity by filling micropores and cracks, thereby increasing cross-linking density and reducing solvent permeability.^{20,21} Excessive inorganic content (e.g., 60 wt %) with insufficient reactive organic intermediates results in uneven distribution of inorganic particles and insufficient covalent bonding, which reduces flexural strength and strain capacity.²² Therefore, to enhance the basic properties and functionality of composite materials, it is essential to investigate both the distribution of inorganic and organic phases and the binding forces between them.^{23,24} When one-dimensional nanomaterials are converted into two-dimensional films through vacuum-assisted filtration or solution casting, lamellar structures of the nanomaterials form under the influence of longitudinal forces (e.g., atmospheric pressure or gravity).²⁵ Fernandes et al. observed that pure cellulose nanocrystals formed a left-handed cholesteric structure during assembly.²⁶ The researchers also noted that the introduction of magnetic fields, alkali solutions, or heterogeneous polymers disrupted this ordered structure.^{27–29} The introduction of supramolecular cross-links (e.g., cucurbit(8)uril-based ternary complexes) enables dynamic reorganization of interfacial interactions under stress, where the dissociation of host–guest complexes promotes localized structural reinforcement, analogous to a brick-and-mortar arrangement, thereby improving the mechanical resilience of the composite.³⁰ Therefore, when studying the enhancement of CSW by HEC, it is crucial to monitor changes in the internal structure of CSW/HEC composite films and the binding forces between the two materials.

One-dimensional inorganic nanocomposites, with their inherent fire resistance and flame retardancy, offer significant advantages for applications in flexible thermoelectric conversion. Silver nanowires, as one-dimensional metallic nanomaterials with high conductivity and efficient thermoelectric conversion, are widely employed in transparent heaters.³¹ The Fourier transform infrared spectroscopy (FTIR) and X-ray photoelectron spectroscopy (XPS) analyses of the polythiophene-silver nanocomposite revealed that the bis-(trifluoromethylsulfonyl)imide (TF₂N) anions interacted with the polymer matrix through specific vibrational modes (e.g., trifluoromethyl group (CF₃) and sulfur dioxide (SO₂) groups) and chemical bonds (e.g., silver–sulfur (Ag–S)), which enhanced the dispersion and interfacial stability of silver nanoparticles within the composite.³² This interfacial interaction was further confirmed by scanning electron microscope (SEM) and transmission electron microscope (TEM) images showing uniformly distributed spherical Ag nanoparticles (10–90 nm) embedded in the polymer matrix, demonstrating

robust structural integrity under mechanical stress. Additionally, the catalytic tests indicated that the composite maintained over 95% efficiency after five cycles, highlighting the durability of the conductive network under operational conditions.

This study utilizes HEC to bond CSW in the preparation of high-temperature-resistant composite materials. The mechanical properties of the composites were evaluated by using a universal tensile testing machine. The internal structure of the composites was examined through scanning electron microscopy (SEM), while the binding properties between the HEC and CSW were analyzed via MD simulations. These analyses assessed the effect of the layered structure and binding forces on the overall properties of the composites. Additionally, a Janus film was fabricated by incorporating silver nanowires (AgNWs) as a conductive layer with CSW/HEC serving as the matrix. This composite film exhibited enhanced thermoelectric performance, and the resulting flexible thermoelectric conversion device demonstrated a high flame retardancy and temperature resistance.

2. EXPERIMENTAL SECTION

2.1. Materials. HEC was provided by Shanghai Macklin Biochemical Technology Co., Ltd. Anhydrous calcium chloride and anhydrous ethanol were purchased from Xilong Scientific Co., Ltd. Sulfuric acid was provided by Chongqing Chuandong Chemical (Group) Co., Ltd. Cetyltrimethylammonium bromide (CTAB) was purchased by Shanghai Yien Chemical Technology Co., Ltd. All other reagents were used as received without further purification.

2.2. Preparation of Calcium Sulfate Whisker. A 0.75 g portion of calcium chloride was dissolved in 50 mL of deionized water, and then 0.02 g of CTAB, 10 mL of anhydrous ethanol, and 0.3 mL of sulfuric acid were added to react at 25 °C for 30 min. After vacuum filtration, a solid filter slag was obtained. After freeze-drying for 25 h, calcium sulfate nanocrystalline whisker was obtained.³³

2.3. Preparation of CSW/HEC Films and CH/AgNWs Films. The preparation of the CSW/HEC films is as follows. First, 0.04 g of HEC is mixed with 3.96 g of deionized water and stirred at 25 °C for 2 h. Then, 0.36 g of CSW and 8.032 g of anhydrous ethanol were mixed and ultrasonically dispersed for 4 min. Then the CSW/anhydrous ethanol mixture and 0.002 g of glycerol were added to the HEC solution and stirred at 25 °C for 6 h.

The mixture was poured into a polytetrafluoroethylene mold (diameter 6 cm) and dried at 45 °C for 12 h.³⁴ Finally, the CSW/HEC composite film was obtained by removing the composite film with an abrasive tool. CSW/HEC15, CSW/HEC20, CSW/HEC25, and CSW/HEC30 composite films were prepared by similar technological processes. Table 1 shows the composition of composite films with different contents.

The preparation of calcium sulfate whiskers/hydroxyethyl cellulose/silver nanowires (CH/AgNWs) films is as follows. First, 1.875 mL of AgNWs (2 mg/mL) was ultrasonically dispersed for 30 s, and the AgNWs films were obtained after vacuum filtration. The CSW/HEC20 mixture was cast on the surface of AgNWs film and dried at 40 °C for 24 h. After the film was uncovered, CH/AgNWs1.5

Table 1. Composition of CSW/HEC Composite Films with Different Contents

sample	CSW (g)	HEC (g)	deionized water (g)	anhydrous ethanol (g)	glycerol (g)
CSW/HEC10	0.36	0.04	3.96	8.03	0.002
CSW/HEC15	0.34	0.06	5.94	6.47	0.003
CSW/HEC20	0.32	0.08	7.92	4.91	0.004
CSW/HEC25	0.30	0.10	9.90	3.35	0.005
CSW/HEC30	0.28	0.12	11.88	1.78	0.006

was obtained. CH/AgNWs2.5 and CH/AgNWs3.5 were prepared by similar process flows (Table 2).

Table 2. Composition of CH/AgNWs Composite Films with Different Contents

sample	CSW (%)	HEC (%)	AgNWs (%)
CH/AgNWs1.5	78.8	19.7	1.5
CH/AgNWs1.5	78.0	19.5	2.5
CH/AgNWs1.5	77.2	19.3	3.5

2.4. Characterizations. The chemical structures of the HEC, CSW, CSW/HEC, and CH/AgNWs films were measured by Fourier transform infrared spectroscopy (Thermo Fisher Scientific Nicolet iS20, America), X-ray diffraction (XRD, Rigaku Ultima IV, Japan), and X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha, America). The morphologies of CSW, CSW/HEC, and CH/AgNWs films were observed using an optical microscope (Nikon Optiphot, Japan) and a scanning electron microscope (SEM; TESCAN MIRA LMS, Czech Republic), and the dimension of the CSW taken was measured using the laser particle size analyzer (Malvern Zetasizer Nano ZS90, Britain). The ^1H chemical shifts of HEC were determined at 400.13 MHz (Thermo Scientific K-Alpha, Germany). The mechanical properties of the films were measured following the GB/T 1040.3-2006 standard using a tensile strength tester (AI-7000-NGD, Goodtechwill, China).

2.5. Molecular Dynamics Simulation. MD simulations was employed to investigate the interaction between HEC and CSW by using the GROMACS 2019.4 software package.³⁵ The chemical formula for HEC is $\text{H}-(\text{C}_6\text{H}_3\text{O}_5)_n-\text{CH}_2\text{CH}_2\text{OH}$, where n is the number of repeat units. Each of the HEC monomer chains contains 10 repeat units, and the geometrical structure of HEC is displayed in the Results. The CSW slab surface model was constructed by CHARMM-GUI with the most stable (010) crystal face applied (Results).^{36,37} The Charmm36m force field was used to describe the interaction between HEC molecules and CSW surface.³⁸ The CSW surface is placed in the center of the simulation box with dimensions of $6.89 \times 6.24 \times 16.00 \text{ nm}^3$. To investigate the effect of the content of HEC molecules on the performance of CSW, 10, 15, 20, 25, and 30 HEC monomer molecules were filled into the simulation box containing the CSW surface, respectively. All systems begin with energy minimization to remove the bad contact between atoms until the maximum force between atoms is lower than $100 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{\AA}^{-1}$. Then, the system is gradually heated to 300.15 K under the NVT (N, number of particles; V, volume; T, temperature) ensemble, which lasts for 1 ns with the V-rescale thermostat.³⁹ After reaching the expected temperature, the subsequent equilibrium is carried out in the NPT (P, pressure) ensemble for 1 ns to maintain 1 bar of pressure via the Berendsen pressure coupling algorithm.⁴⁰ Finally, for each system, a 100 ns MD simulation was performed without any restrictions under the NVT ensemble with a time step of 2 fs. The Lennard-Jones and electrostatic interactions were truncated at a cutoff distance of 1.2 nm. The long-ranged electrostatic terms were evaluated using the particle

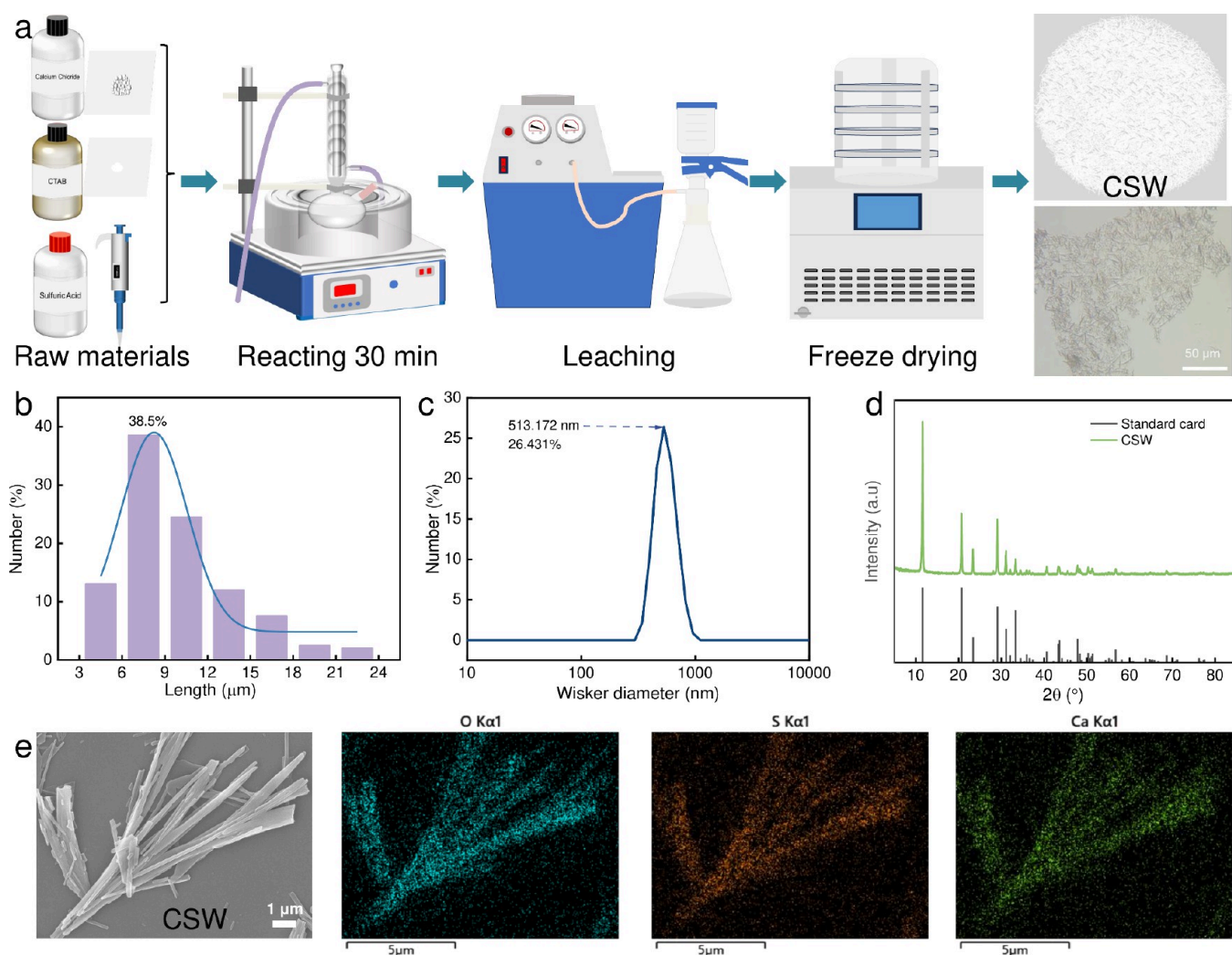


Figure 1. Preparation process (a), length distribution (b), diameter distribution (c), and XRD (d) SEM and EDs (e) of CSW.

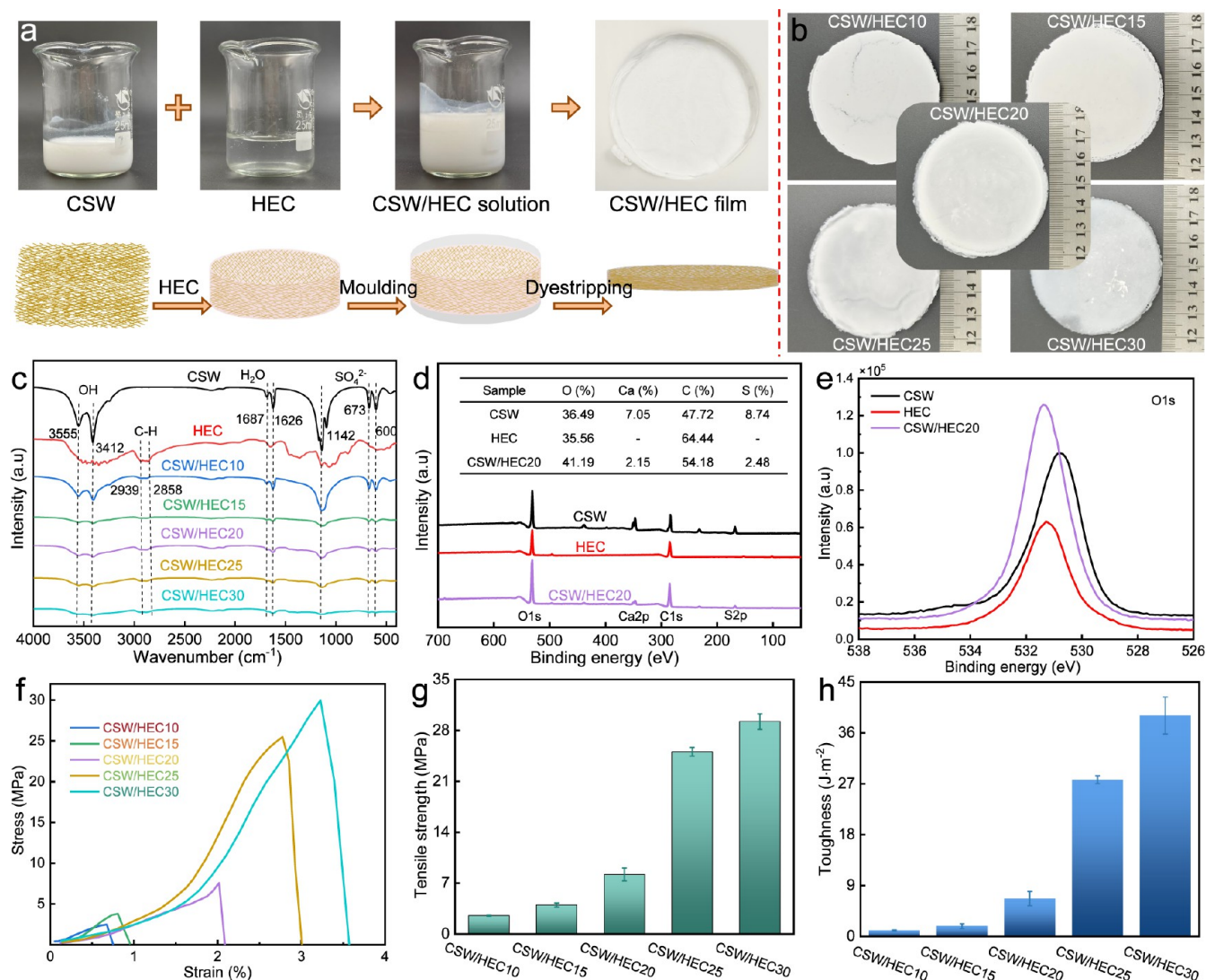


Figure 2. Preparation process (a), optical photographs (b), FTIR (c), XRD (d), XPS (e), stress–strain (f), tensile strength (g), and toughness (h) of CSW/HEC film.

mesh Ewald summation.⁴¹ Bond lengths with hydrogen were constrained with the linear constraint solver (LINCS) algorithm.⁴² Periodic boundary conditions were applied in all three dimensions. The trajectories were visualized via the visual molecular dynamics (VMD) 1.9.3 software.⁴³

2.6. Temperature Resistance and Flame Retardant Test. The heat treatment of the film was carried out by a heating table, and the heat resistance temperature of the film was recorded by a thermal imager (Nuofangzhou NF-583s, China).¹⁵ The limiting oxygen index (LOI) of HEC and CSW/HEC films was tested by an oxygen index tester (Dimosi COI, China) according to GB/T2406-1993.

2.7. Electric Heating Test. The temperature of the CH/AgNWs film AgNWs layer was measured by a thermocouple with adjustable direct current power supply, and the temperature change data were recorded by the corresponding driving software.¹⁴

3. RESULTS AND DISCUSSION

Figure 1a illustrates the fabrication process of the CSW. CSW was synthesized via liquid-phase synthesis using calcium chloride and sulfuric acid. The CSW powder was then collected through filtration and freeze-drying. The length of the CSW ranges from 3 to 24 μm (Figure 1b), with the predominant distribution between 6 and 9 μm . The diameter

of the CSW is most frequently observed at 513 nm (Figure 1c). The CSW powder was confirmed by XRD analysis (Figure 1d).⁴⁴ Scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) revealed that CSW has a one-dimensional linear structure and is composed of oxygen (O), sulfur (S), and calcium (Ca) elements (Figure 1e). The degree of hydroxyethyl substitution in the HEC molecule was found to be 2.57 (Figure S1).

As shown in Figure 2a, CSW/HEC films were prepared through solution evaporation of the mixed dispersions. Figure 2b presents optical images of the white CSW/HEC films. FTIR spectroscopy was employed to characterize the composition of the CSW, HEC, and CSW/HEC films (Figure 2c). The absorption bands at 1142, 673, and 600 cm^{-1} are attributed to SO_4^{2-} ions in the CSW sample.⁴⁵ Additionally, the absorption bands at 3412 and 3555 cm^{-1} correspond to the O–H vibration of sulfuric acid and the stretching vibration of O–H bonds in water molecules, respectively. The peaks around 2939 and 2858 cm^{-1} in HEC are due to CH_3 – stretching vibrations. The intensities of the characteristic CSW peaks decrease with an increasing HEC content. The significantly broadened absorption band spanning 3000–3700 cm^{-1} further confirms

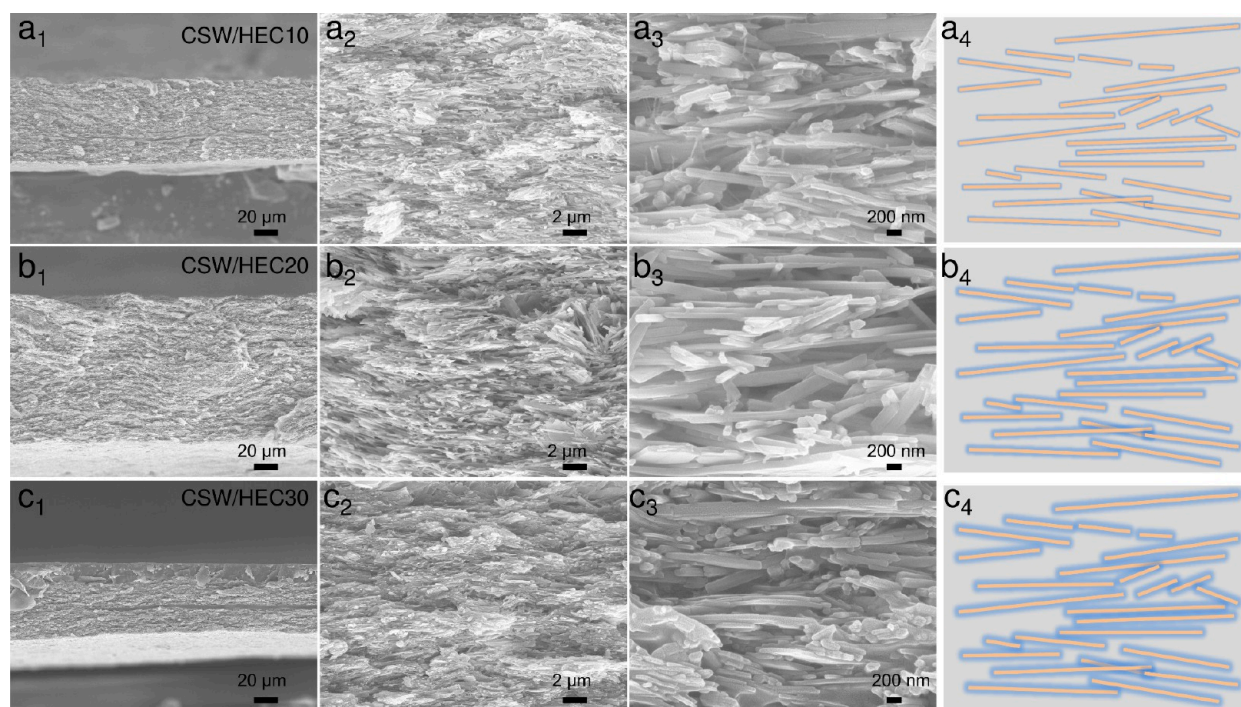


Figure 3. Cross-section SEM image of CSW/HEC10 (a), CSW/HEC20 (b), and CSW/HEC30 (c) films at different scales.

hydrogen bonding interactions between HEC and CSW at their interface. Figures 2d and 2e show the XPS spectra of the composite film. Compared to HEC, CSW contains 7.05% Ca and 8.74% O elements. However, the content of Ca and S elements in CSW/HEC20 films decreased to 2.15% and 2.48%, respectively. The glycosidic bond in HEC exhibits a higher binding energy at ~ 533.5 eV, whereas the lattice oxygen of CSW appears at ~ 530.5 eV. Consequently, the O 1s spectrum of the CSW/HEC composite demonstrates a left-shifted chemical displacement due to the dominant contribution from CSW's low-energy oxygen species.⁴⁶

Figures 2f–2h present the mechanical properties of the composite films. As shown in the stress–strain diagram, when the content of HEC ranges from 10% to 30%, the mechanical properties of the composite films increase with increasing HEC content. The tensile strength and toughness of CSW/HEC20 are 8.20 MPa and $6.68 \text{ J}\cdot\text{m}^{-2}$, respectively, representing increases of 224% and 530% compared to those of CSW/HEC10. This improvement is attributed to the bonding effect of the polymer HEC on the inorganic materials. When the HEC content was further increased to 25%, the tensile strength and toughness of CSW/HEC25 rose to 25.07 MPa and $27.72 \text{ J}\cdot\text{m}^{-2}$, which are 206% and 315% higher than those of CSW/HEC20, respectively. This enhancement is likely due to the network structure formed by HEC itself.¹⁶ However, the tensile strength and toughness of CSW/HEC30 increased by only 16% and 41%, respectively, compared to CSW/HEC25. Collectively, CSW/HEC10, CSW/HE15, and CSW/HE20 specimens exhibited brittle fracture behavior, with fracture strains below 2.5%. In contrast, CSW/HEC25 and CSW/HE30 films demonstrated ductile failure, where the HEC matrix absorbed substantial energy through molecular chain disentanglement and hydrogen bond reorganization, thereby enhancing overall toughness.⁴⁷

We investigated the internal structures of the composite films and constructed a model diagram to analyze the

mechanisms through which HEC enhances the mechanical properties of CSW and improves the temperature resistance and flame-retardant performance of CSW/HEC films (Figure 3). In Figure 3a, the CSW whiskers in CSW/HEC10 films are uniformly dispersed with no apparent interconnection between them. When the HEC content was increased to 20%, a distinct HEC coating layer appeared on the CSW surface and the whiskers were interconnected by HEC (Figure 3b). Organic HEC can effectively transfer stress to inorganic CSW, thereby enhancing the mechanical properties of the material.⁴⁸ Similarly, in HEC/BNNS composites, connections between BNNS were formed when the BNNS content reached 20%.¹⁵ In Figure 3c, a large amount of HEC polymer is observed on the CSW surface with some cracking of the HEC under the electron beam of the scanning electron microscope (SEM). The SEM and EDS images of CSW/HEC composite films are shown in Figure S2, where it can be observed that S and Ca elements are uniformly distributed, indicating that the CSW is well-dispersed in the system. The C element primarily originates from the HEC molecules (Figure S3), suggesting that HEC is dispersed in the gaps between the CSW whiskers, effectively linking them.

To further elucidate the interaction mechanism between the HEC and CSW in CSW/HEC composite films at the molecular level, MD simulations were performed. These simulations help to explain the differences in the mechanical properties of various composite materials. As can be seen from Figures 4a and Table S1, after the MD simulation has been carried out for approximately 60 ns, the total interaction energy (E_{tot}) between HEC and CSW remains almost unchanged, which indicates that the simulation has reached an equilibrium state. The interaction energy between HEC and CSW initially increases and then decreases with an increasing HEC content. The E_{tot} between CSW and HEC is maximized at 20% HEC, with a value of -9227.23 kJ/mol . The interaction energy between nanocellulose and cement matrix composites

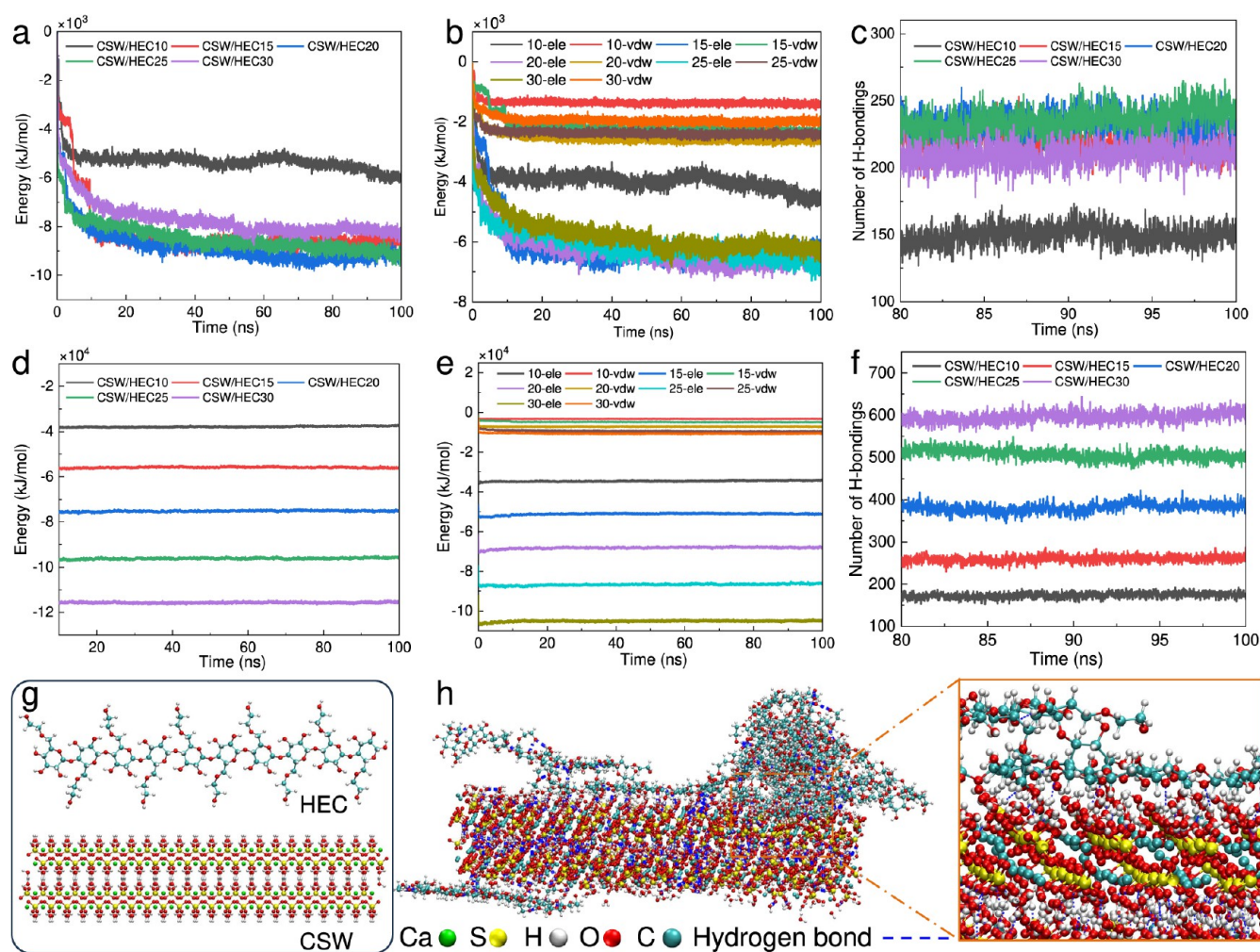


Figure 4. (a) Time evolution of E_{tot} between CSW and HEC, (b) time evolution of E_{ele} and E_{vdw} between CSW and HEC, (c) the number of H-bonds between the CSW and HEC, (d) time evolution of E_{tot} between HEC and HEC, (e) time evolution of E_{ele} and E_{vdw} between HEC and HEC, (f) the number of H-bonds between the HEC and HEC, (g) structure model of HEC and CSW, and (h) snapshot of H-bond interaction pattern between CSW and HEC and HEC and HEC.

was $-26,945$ kJ/mol, which is larger than the interaction between CSW and HEC.⁴⁹ Notably, the electrostatic interaction (E_{ele}) is the primary force driving the binding of CSW and HEC, while the van der Waals interaction (E_{vdw}) plays a complementary role, with values of -6618.10 and -2609.13 kJ/mol, respectively (Figure 4b). The number of hydrogen bonds (H-bonds) between HEC and CSW was also analyzed to gain further insight into the interaction process, as shown in Figure 4c. As can be seen from Figure 4h, the oxygen atoms in the hydroxyl groups on the HEC and the hydrogen atoms on the crystalline water in the CSW can form hydrogen bonds.

The results indicate that the CSW/HEC25 sample exhibits the highest number of H-bonds, with a value of 235.35, which is approximately 85 more pairs of H-bonds than that of the CSW/HEC10 sample. This observation is consistent with the tensile strength results. When the HEC content was increased from 25% to 30%, the E_{tot} between CSW and HEC in CSW/HEC30 decreased to -8147.62 kJ/mol, and the number of H-bonds dropped to 209.31. Despite this, the mechanical properties of the CSW/HEC composite films continued to improve, primarily due to the presence of strong bonding forces not only between CSW and HEC but also within the

HEC network itself. The interaction energy between HEC molecules increases with higher HEC content, with the E_{tot} reaching its peak value of $-115,551.88$ kJ/mol at 30% HEC (Figure 4d). Moreover, the number of H-bonds between HEC molecules rose to 596.14, surpassing the number of hydrogen bond interactions between the CSW and HEC (Figure 4f). Consequently, the CSW/HEC samples with 30% HEC exhibited the highest tensile strength and toughness.

The temperature resistance of various CSW/HEC films was tested by using a heating table, as shown in Figures 5a–5c. After treatment at 105 °C for 9 min, the morphology of the composite films with HEC contents ranging from 10% to 20% remained relatively unchanged. However, the edges of the CSW/HEC25 and CSW/HEC30 films deteriorated after just 3 min of treatment, with further damage observed after 9 min of exposure. It was found that the temperature resistance of the composite films decreased as the HEC content increased (Figures 5d and 5e). Notably, the CSW/HEC20 film retained a good structure at temperatures up to 320 °C. Li et al. combined pulping with CSW to prepare paper-based materials, achieving a temperature resistance greater than 130 °C.⁵⁰ Figures 5f and 5g show the flame retardancy of the different films. Materials with a limiting oxygen index of $>27\%$ are

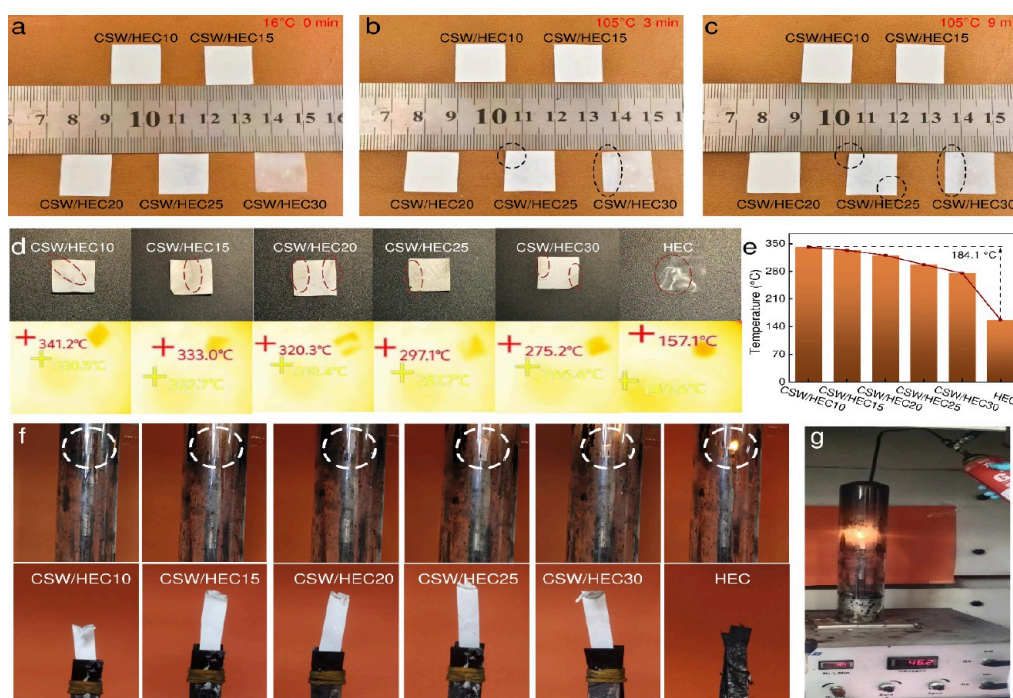


Figure 5. Optical images of the CSW/HEC films (a) at 16 °C, (b) after heating to 3 min at 105 °C, and (c) after treating to 9 min at 6 kPa and 105 °C. Temperature-resistant photos (d) and temperature curves (e) of the CSW/HEC films. Flame retardant rendering (f) of CSW/HEC films and (g) of the burner used for flame retardant tests in 44.5% O₂ concentration.

considered flame-retardant materials. The results indicate that all CSW/HEC films exhibit excellent self-extinguishing performance with a limiting oxygen index (LOI) greater than 44.5%. However, the HEC itself exhibits significant combustion characteristics.

The CSW/HEC20 film exhibits excellent temperature resistance and mechanical properties. Janus composite films were fabricated by using CSW/HEC20 as the insulating substrate and AgNWs as the conductive layers (Figure 6a). The AgNWs have a diameter of approximately 250 nm and a length of 10 μ m (Figure S4). The CH side of the Janus film appears white (Figure 6b), while the AgNWs side is dark gray (Figure 6c). Figures 6d–6i present the SEM and EDS cross-sectional images of the composite film. The homogeneous distribution of S and Ca elements indicates that CSW is uniformly dispersed throughout the system. The C and O elements reveal that the AgNWs are primarily fixed on the surface of the CH film via HEC. The distribution of Ag suggests that the AgNWs are embedded within the CH film.

Figure 7 presents the SEM images of Janus films with varying AgNWs content. In the CH/AgNWs1.5 film, AgNWs are embedded on the surface of the CH film and the AgNWs, HEC, and CSW are uniformly distributed (Figure 7a). However, it is evident that the AgNWs are not fully covered across the film's surface. In the CH/AgNWs2.5 film, a dense network of AgNWs forms covering the surface of the CH film (Figure 7b). A similar phenomenon was observed when hydroxypropyl cellulose was loaded with AgNWs.³³ In the CH/AgNWs3.5 sample, a conductive layer approximately 4 μ m thick is formed by the AgNWs, and layers of overlapping CSW and AgNWs are tightly connected by an HEC (Figure 7c).

The Janus composite film was applied in electrothermal conversion, and its performance was evaluated by using a thermocouple and a thermal infrared camera (Figure 8a). As

the AgNWs content increased, the resistance of the conductive layer gradually decreased (Figure 8b). The highest temperature achieved by the CH/AgNWs1.5 film was 86.5 °C, while the CH/AgNWs2.5 and CH/AgNWs3.5 films reached maximum temperatures of 78.0 and 63.5 °C, respectively. Thermal infrared imaging further revealed that the composite film attained its highest temperature within 20 s, demonstrating rapid heating performance. The relatively low temperature of the CH/AgNWs3.5 composite film may be attributed to the high thermal conductivity of the AgNWs. The CSW-based Janus composite film can reach a temperature of 85.0 °C within 10 s under a 0.6 V voltage, making it promising for use in flexible electric heaters. When utilizing HEC/boron nitride nanosheets as the substrate for AgNWs, the maximum heating temperature attains only 53.1 °C.⁵¹ In contrast, the CH/AgNWs1.5 composite markedly surpasses the performance of conventional cellulose-derived heating systems.

4. CONCLUSIONS

This study successfully prepared a CSW/HEC composite material with high temperature resistance and flame retardancy using a solution casting method. A Janus composite film with electrothermal conversion capabilities was also fabricated by incorporating silver nanowires. The tensile strength of the CSW/HEC20 composite film is 8.20 MPa, with a limiting oxygen index greater than 44.5%, classifying it as a non-combustible material. MD simulations revealed that the electrostatic and van der Waals interactions between HEC and CSW in the CSW/HEC20 sample are the strongest, with a total interaction energy of 9227.23 kJ/mol. Additionally, strong forces exist between the HEC molecules themselves. The CSW-based Janus composite film can reach a temperature of 85.0 °C within 10 s at 0.6 V, showing promising potential for use in flexible electric heaters.

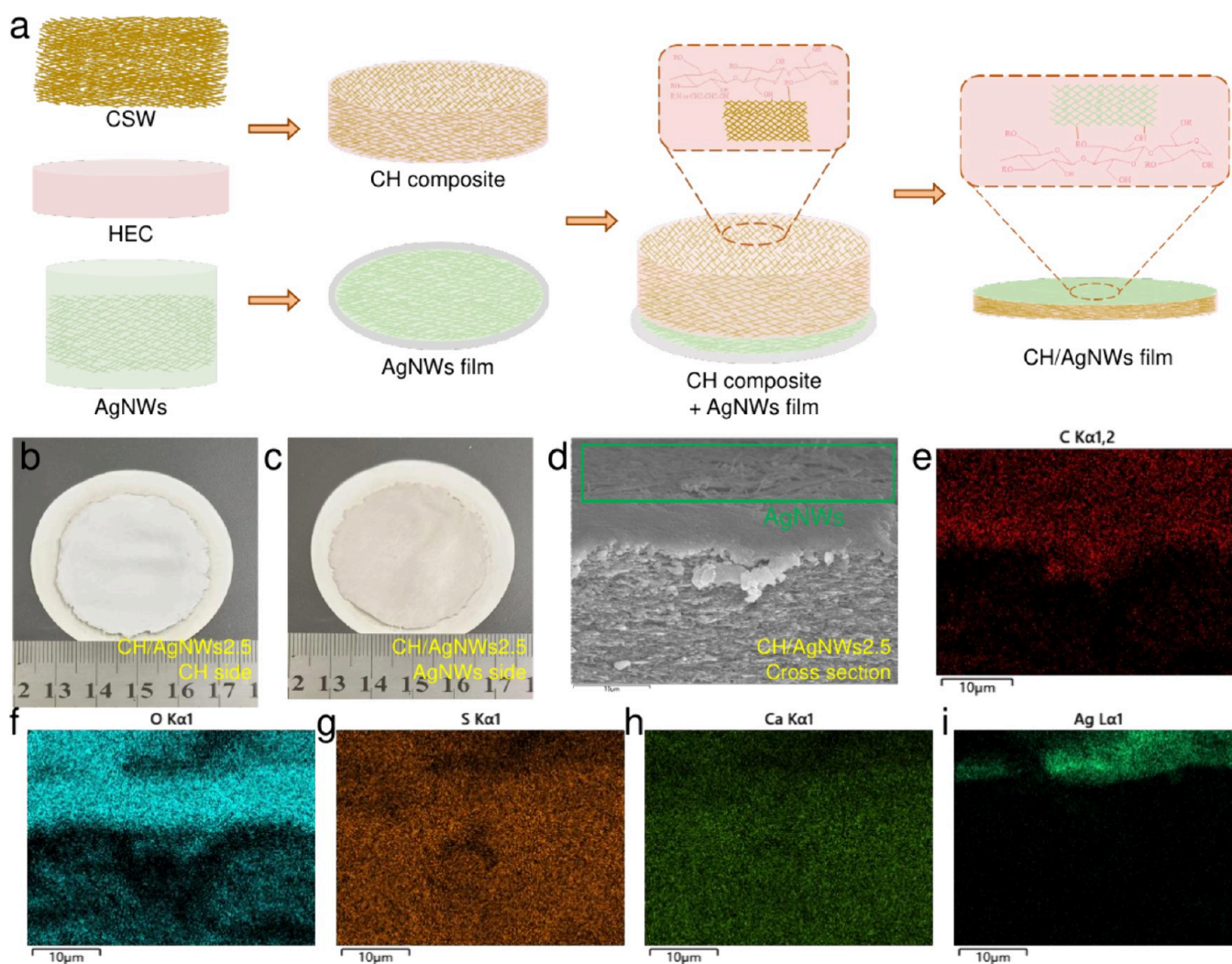


Figure 6. Preparation drawing (a) and optical photo (b and c) of CH/AgNWs film. (d–i) Cross-sectional morphologies and EDS of CH/AgNWs2.5.

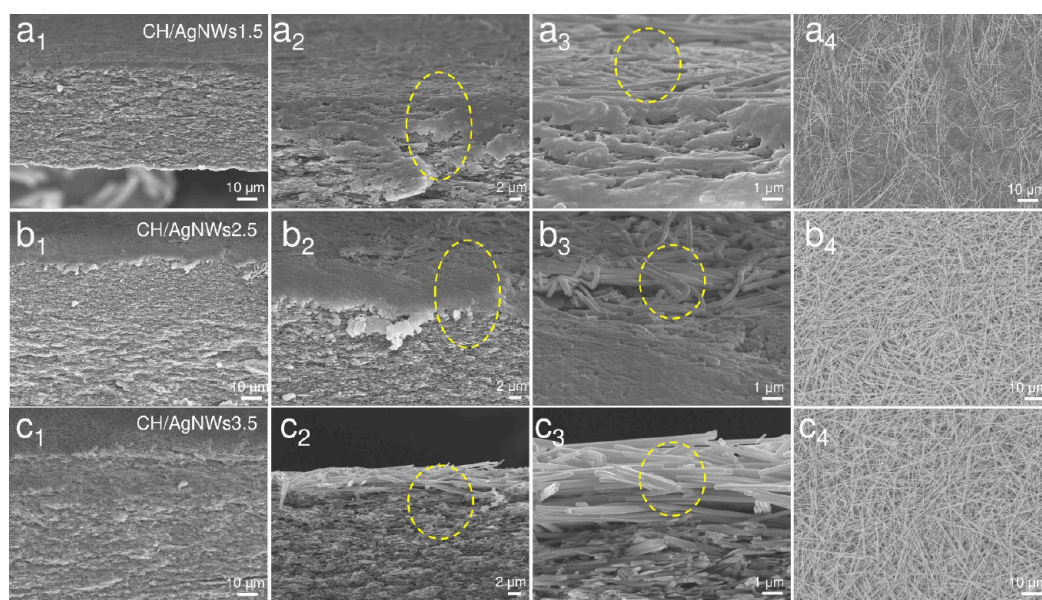


Figure 7. Cross-sectional morphologies and planar images of (a) CH/AgNWs1.5, (b) CH/AgNWs2.5, and (c) CH/AgNWs3.5 films on different scales.

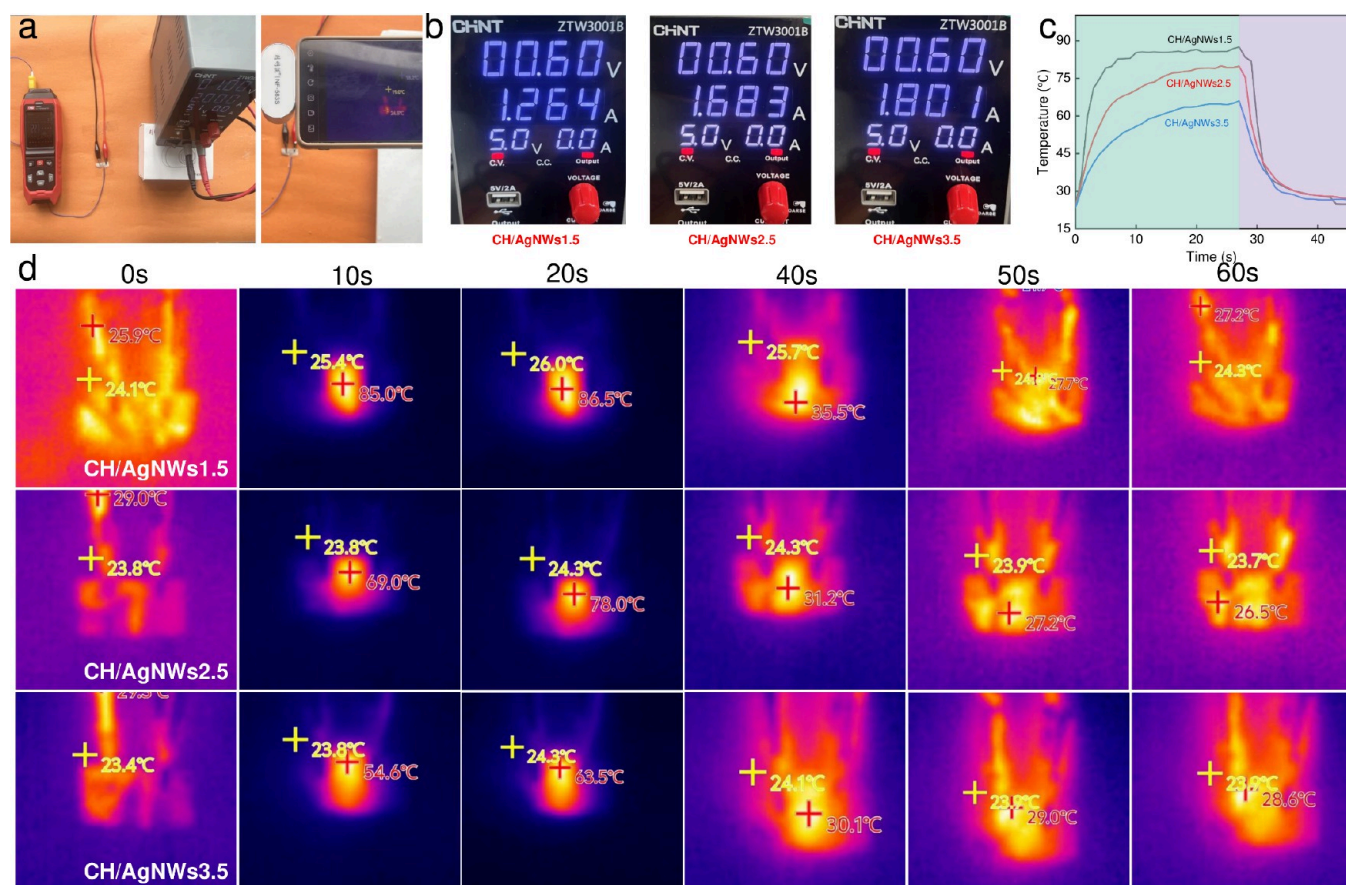


Figure 8. Test equipment (a), current and voltage photos (b), and temperature–time curve (c) of the CH/AgNWs films. (d) Thermal diffusion images of the CH/AgNWs1.5, CH/AgNWs2.5, and CH/AgNWs3.5 films at 0.6 V voltage.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.5c01925>.

Typical solution state ^1H NMR spectrum of the HEC; cross-section SEM image and EDS of CSW/HEC films; cross-section SEM image and EDS of HEC film; TEM image of the AgNWs (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Zhiyong Xu — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China; Email: ZYXu@kust.edu.cn

Yuxin Liu — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China; orcid.org/0000-0002-9850-8413; Email: yuxinliu@kmust.edu.cn

Authors

Jizhen Huang — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China; orcid.org/0000-0002-1618-417X

Shaolu Qi — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China

Jiao Ji — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China

Wenbo Zhao — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China; orcid.org/0000-0003-3119-9342

Jianquan Hu — Faculty of Chemical Engineering, Kunming University of Science and Technology, Kunming 650500, China; orcid.org/0000-0002-9618-8997

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsapm.5c01925>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (22468028), the Yunnan Fundamental Research Projects (202501AU070141), the Analysis and Testing Foundation of Kunming University of Science and Technology (2024T20230145), and the project supported by "Xing Dian Ying Cai" programme of Yunnan province.

■ REFERENCES

- (1) Jang, H.; Ahn, J.; Jeong, Y.; Ha, J. H.; Jeong, J. H.; Oh, M. W.; et al. Flexible all-inorganic thermoelectric yarns. *Adv. Mater.* **2024**, *36* (47), 2408320.
- (2) Chen, W.; Shi, X. L.; Li, M.; Liu, T.; Mao, Y.; Liu, Q.; et al. Nanobinders advance screen-printed flexible thermoelectrics. *Science* **2024**, *386*, 1265–71.

- (3) He, Y.; Xu, X.; Xiao, S.; Wu, J.; Zhou, P.; Chen, L.; et al. Research progress and application of multimodal flexible sensors for electronic skin. *ACS Sens* **2024**, 9 (5), 2275–93.
- (4) Zhou, Y.; Zhang, Y.; Nie, Y.; Sun, D.; Wu, D.; Ban, L.; et al. Recent advances and perspectives in functional chitosan-based composites for environmental remediation, energy, and biomedical applications. *Progress in Materials Science* **2025**, 152, 101460.
- (5) Nie, Y.; Zhou, Y.; Zhang, Y.; Sun, D.; Wu, D.; Ban, L.; et al. Sustainable synthesis of functional materials assisted by deep eutectic solvents for biomedical, environmental, and energy applications. *Adv. Funct. Mater.* **2025**, 35, 2418957.
- (6) Ahn, J.; Jeong, Y.; Kang, M.; Ahn, J.; Padmajan Sasikala, S.; Yang, L.; et al. Nanoribbon yarn with versatile inorganic materials. *Small* **2024**, 20 (38), 2311736.
- (7) Kang, T. W.; Kim, S. H.; Kim, C. H.; Lee, S. M.; Kim, H. K.; Park, J. S.; et al. Flexible polymer/metal/polymer and polymer/metal/inorganic trilayer transparent conducting thin film heaters with highly hydrophobic surface. *ACS Appl. Mater. Interfaces* **2017**, 9 (38), 33129–36.
- (8) Yuan, W.; Cui, J.; Xu, S. Mechanical properties and interfacial interaction of modified calcium sulfate whisker/poly(vinyl chloride) composites. *J. Mater. Sci. Technol.* **2016**, 32 (12), 1352–60.
- (9) Wang, X.; Fu, J.; Zou, F.; Luo, H.; Cai, J.; Yu, Y. High-Performance Composite Aerogels with CaSO₄ Whisker and Nanofiller Synthesized from Lepidolite Lithium Slag. *ACS Appl. Polym. Mater.* **2024**, 6 (24), 15058–15069.
- (10) Lu, Y.; Jiang, N.; Li, X.; Xu, S. Effect of inorganic-organic surface modification of calcium sulfate whiskers on mechanical and thermal properties of calcium sulfate whisker/poly(vinyl chloride) composites. *RSC Adv.* **2017**, 7, 46486–46498.
- (11) Aakriti; Maiti, S.; Jain, N.; Prajapati, P. Synthesis of calcium sulfate whiskers via acidification exploiting FGD gypsum for improved binder properties. *Sustain. Chem. Pharm.* **2024**, 42, 101745.
- (12) Zhang, Y. M.; Wu, J.; Lu, Q.; Qin, S. C.; Liu, Z. G.; Hu, Y. H.; et al. Enhancing mechanical properties of pvc composites through surface composite modified calcium sulfate whiskers. *Mater. Sci. Poland* **2024**, 42 (3), 55–71.
- (13) Cong, S.; Lan, T.; Wang, Y.; Zu, L.; Dong, S.; Zhang, Z.; et al. Titanium dioxide and calcium sulfate whiskers are used for the preparation of high performance polypropylene and reduce white pollution. *Langmuir* **2024**, 40 (21), 11056–66.
- (14) Huang, J.; Hua, L.; Li, J.; Xu, X.; Song, L.; Lu, Z. Sandwiched film of graphene/silver nanowire conductive layer reinforced by hydroxyethyl cellulose bond layer. *Int. J. Biol. Macromol.* **2024**, 258, 128883.
- (15) Huang, J.; E, S.; Li, J.; Jia, F.; Ma, Q.; Hua, L.; et al. Ball-milling exfoliation of hexagonal boron nitride in viscous hydroxyethyl cellulose for producing nanosheet films as thermal interface materials. *ACS Appl. Nano Mater.* **2021**, 4 (12), 13167–75.
- (16) Huang, J.; E, S.; Si, L.; Li, J.; Tian, Z.; Lu, Z. Composite films of hydroxyethyl cellulose and hydroxyapatite nanowires with high mechanical strength and electrical insulation property. *J. Wood Chem. Technol.* **2022**, 42 (1), 15–25.
- (17) Wang, N.; Liu, H.; Sun, M.; et al. Achieving Wide-Temperature-Range Sustainable Zinc-Ion Batteries via Magnesium-Doped Cathodes and Gel Electrolytes. *ACS Sustainable Chem. Eng.* **2024**, 12 (9), 3527–37.
- (18) Marti, D.; Petuya, R.; Bosoni, E.; Dublanchet, A.-C.; Mohr, S.; Leonforte, F. Predicting the glass transition temperature of biopolymers via high-throughput molecular dynamics simulations and machine learning. *ACS Appl. Polym. Mater.* **2024**, 6 (8), 4449–4461.
- (19) Lin, Z.; Liu, H.; Richardson, J. J.; Xu, W.; Chen, J.; Zhou, J.; et al. Metal-phenolic network composites: from fundamentals to applications. *Chem. Soc. Rev.* **2024**, 53 (22), 10800–26.
- (20) Fu, Y.; Gu, Z.; Gan, Q.; Mai, Y.-W. A review on the ionic conductivity and mechanical properties of composite polymer electrolytes (CPEs) for lithium batteries: Insights from the perspective of polymer/filler composites. *Mat. Sci. Eng. R* **2024**, 160, 100815.
- (21) Hazarika, D.; Srivastava, R. K.; Hakkarainen, M. Light processable methacrylated carbon dot-Hydroxyethyl cellulose resins with potential applications from dye adsorption to antibacterial gels and wet wipes. *ACS Appl. Polym. Mater.* **2024**, 6 (11), 6776–6787.
- (22) Das, D.; Bhattacharjee, S.; Bhaladhare, S. Preparation of Cellulose Hydrogels and Hydrogel Nanocomposites Reinforced by Crystalline Cellulose Nanofibers (CNFs) as a Water Reservoir for Agriculture Use. *ACS Appl. Polym. Mater.* **2023**, 5 (4), 2895–2904.
- (23) Lubineau, G.; Alfano, M.; Tao, R.; Wagih, A.; Yudhanto, A.; Li, X.; et al. Harnessing extrinsic dissipation to enhance the toughness of composites and composite joints: a state-of-the-art review of recent advances. *Adv. Mater.* **2024**, 36 (S1), 2407132.
- (24) Wu, J.; Wang, Y.; Zhang, J.; Zhao, C.; Fan, Z.; Shu, Q.; et al. A lightweight aramid-based structural composite with ultralow thermal conductivity and high-impact force dissipation. *Matter* **2022**, 5 (7), 2265–84.
- (25) He, M.; Huang, Y.; Zhang, X.; Zhu, W.; Shao, W.; Wang, J.; et al. Flexible cellulose nanofibers/MXene composite films for UV-shielding packaging. *Int. J. Biol. Macromol.* **2024**, 264, 130821.
- (26) Fernandes, S. N.; Almeida, P. L.; Monge, N.; Aguirre, L. E.; Reis, D.; de Oliveira, C. L.; et al. Mind the microgap in iridescent cellulose nanocrystal films. *Adv. Mater.* **2017**, 29 (2), 1603560.
- (27) Frka-Petecic, B.; Guidetti, G.; Kamita, G.; Vignolini, S. Controlling the photonic properties of cholesteric cellulose nanocrystal films with magnets. *Adv. Mater.* **2017**, 29 (32), 1701469.
- (28) Nan, F.; Nagarajan, S.; Chen, Y.; Liu, P.; Duan, Y.; Men, Y.; et al. Enhanced toughness and thermal stability of cellulose nanocrystal iridescent films by alkali treatment. *ACS Sustainable Chem. Eng.* **2017**, 5 (10), 8951–8.
- (29) Ge, W.; Wei, Q.; Zhang, F.; Feng, Z.; Bai, X.; Feng, S.; et al. Sensitive chemoselectivity of cellulose nanocrystal films. *Cellulose* **2022**, 29 (7), 4097–107.
- (30) Das, A. D.; Mannoni, G.; Fruh, A. E.; Orsi, D.; Pinalli, R.; Dalcanele, E. Damage-Reporting carbon fiber epoxy composites. *ACS Appl. Polym. Mater.* **2019**, 1 (11), 2990–2997.
- (31) Hua, L.; Song, L.; Chen, W.-J.; Huang, J.; Kou, Y.; Yu, S.; et al. Multifunctional composite flexible films of hydroxypropyl cellulose and silver nanowires. *Ind. Crop Prod* **2024**, 215, 118532.
- (32) Tavares, F.; Oliveira, L. V. F.; Hajjar-Garreau, S.; Camilo, F. F. Synthesis of polythiophene-Silver nanocomposites in ionic liquid for catalytic applications. *ACS Appl. Polym. Mater.* **2024**, 6 (7), 3756–3766.
- (33) Chen, Q.; Jia, C.; Li, Y.; Xu, J.; Guan, B.; Yates, M. Z. Alpha-Calcium sulfate hemihydrate nanorods synthesis: a method for nanoparticle preparation by mesocrystallization. *Langmuir* **2017**, 33 (9), 2362–9.
- (34) Huang, J.; Lu, Z.; Li, J.; Ning, D.; Jin, Z.; Ma, Q.; et al. Improved mechanical and ultraviolet shielding performances of hydroxyethyl cellulose film by using aramid nanofibers as additives. *Carbohydr. Polym.* **2021**, 255, 117330.
- (35) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; et al. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, 1–2, 19–25.
- (36) Jo, S.; Kim, T.; Iyer, V. G.; Im, W. CHARMM-GUI: a web-based graphical user interface for CHARMM. *J. Comput. Chem.* **2008**, 29 (11), 1859–65.
- (37) Choi, Y. K.; Kern, N. R.; Kim, S.; Kanhaiya, K.; Afshar, Y.; Jeon, S. H.; et al. Charmm-gui nanomaterial modeler for modeling and simulation of nanomaterial systems. *J. Chem. Theory Comput* **2022**, 18 (1), 479–93.
- (38) Huang, J.; Rauscher, S.; Nawrocki, G.; Ran, T.; Feig, M.; de Groot, B. L.; et al. CHARMM36m an improved force field for folded and intrinsically disordered proteins. *Nat. Methods* **2017**, 14, 71–73.
- (39) Bussi, G.; Donadio, D.; Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **2007**, 126 (1), 014101.

(40) Berendsen, H. J. C.; Postma, J. P. M.; van Gunsteren, W. F.; DiNola, A.; Haak, J. R. Molecular dynamics with coupling to an external bath. *J. Chem. Phys.* **1984**, *81* (8), 3684–90.

(41) Feller, S. E.; MacKerell, A. D. An improved empirical potential energy function for molecular simulations of phospholipids. *J. Phys. Chem. B* **2000**, *104* (31), 7510–15.

(42) Hess, B.; Bekker, H.; Berendsen, H. J. C.; Fraaije, J. G. E. M. LINCS: A linear constraint solver for molecular simulations. *J. Comput. Chem.* **1997**, *18* (12), 1463–72.

(43) Humphrey, W.; Dalke, A.; Schulten, K. VMD: Visual molecular dynamics william humphrey, andrew dalke, and klaus schulten. *J. Mol. Graph Model* **1996**, *14*, 33–38.

(44) Aakriti; Bhardwaj, A.; Maiti, S.; Jain, N.; Pathak, A.; Gupta, R. R. Calcium sulphate whiskers (CSW) an innovative material for civil engineering applications: A critical review of its preparation, characterization, current trends, and prospects. *Constr Build Mater.* **2024**, *420*, 135624.

(45) Fan, H.; Song, X.; Xu, Y.; Yu, J. Insights into the modification for improving the surface property of calcium sulfate whisker: Experimental and DFT simulation study. *Appl. Surf. Sci.* **2019**, *478*, 594–600.

(46) Zaharani, L.; Vequizo, M. K.; Amiri, Z. R.; Johan, M. R.; Majid, W. H.; Khaligh, N. G. Development of a new sustainable greener strategy for cellulose functionalization: A mechanochemical catalyst-free and solvent-free process in ambient conditions. *Int. J. Biol. Macromol.* **2025**, *304*, 140907.

(47) Fu, Q.; Qin, Y.; Zhang, J.; Guo, W.; Zhang, K.; Xiao, T.; et al. Colored gold ions enabled high-transparent and low-haze cellulose films with excellent flame-retardant and uv to blue light-blocking performance. *ACS Appl. Polym. Mater.* **2025**, *7* (8), 5089–5103.

(48) Jia, F.; Lu, Z.; Huang, T.; Xu, M.; Xu, X.; Guo, Z.; et al. Twin-coated skeleton PEDOT: PSS/MXene/para-aramid nanofibers hybrid aerogel with efficient EMI shielding performance and tunable power coefficient. *Adv. Compos Hybrid Ma* **2025**, *8*, 200.

(49) Zhang, L.; Lin, D.; Zhang, M.; Wang, S.; Zhang, W.; Song, S. Molecular dynamics simulation of nanocellulose and cement matrix composites. *Construction and Building Materials* **2024**, *449*, 138342.

(50) Li, Y.; Liu, Y.; Huang, J.; Mei, Y. Improved length of calcium sulfate crystal seeds and whiskers via ball milling and hydration treatment. *Chinese J. Chem. Eng.* **2024**, *71*, 102–9.

(51) Huang, J.; Hua, L.; Song, L.; Li, J.; E, S.; Lu, Z. Hydroxyethyl cellulose/boron nitride nanosheet films coated with silver nanowires for electronic packaging applications. *Mater. Today Sustain* **2023**, *21*, 100307.



The graphic features a collage of scientific images and text. At the top, it says 'CAS Insights™' and 'Exploring the innovations shaping tomorrow'. Below this, it says 'Discover the latest scientific research and trends with CAS Insights. Subscribe for email updates on new articles, reports, and webinars at the intersection of science and innovation.' A yellow button says 'Subscribe today'. At the bottom right is the CAS logo, which includes the text 'CAS' and 'A division of the American Chemical Society'.

CAS INSIGHTS™
EXPLORE THE INNOVATIONS SHAPING TOMORROW

Discover the latest scientific research and trends with CAS Insights. Subscribe for email updates on new articles, reports, and webinars at the intersection of science and innovation.

Subscribe today

CAS
A division of the American Chemical Society